

Quantum Computing and Machine Learning

Quantum Lecture Series

Department of Quantum Information Science

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Outline

What is Quantum Computing?

Quantum computing leverages principles of quantum mechanics to perform computations beyond classical capabilities.

Key Concepts:

- **Superposition:** Qubits can exist in a combination of states.
- **Entanglement:** Correlation between qubits regardless of distance.
- **Quantum Interference:** Probability amplitudes interfere to solve problems.

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Qubit Representation:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1$$

What is Machine Learning?

Machine Learning (ML) is the study of algorithms that improve through data experience.

Types of Machine Learning:

- **Supervised Learning:** Labeled data for classification or regression.
- **Unsupervised Learning:** No labels; discover hidden patterns.
- **Reinforcement Learning:** Learning through interaction with the environment.

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ML Workflow:

Data → Model Training → Prediction

What is Quantum Machine Learning?

Quantum Machine Learning (QML) integrates quantum computing with machine learning algorithms to exploit quantum advantages.

Motivation:

- High-dimensional Hilbert spaces for better feature representation.
- Quantum parallelism for faster computation.
- Quantum entanglement for richer data encoding.

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Quantum Model Example:

$U(\theta) |x\rangle \implies$ Quantum Kernel for Classification

1. Quantum Support Vector Machines (QSVM)

Quantum Kernel Estimation:

- Maps classical data to a quantum Hilbert space.
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Quantum Kernel:

$$K(x, x') = |\langle \psi(x) | \psi(x') \rangle|^2$$

Advantage: - Potentially exponential speedup over classical SVMs.

2. Quantum Neural Networks (QNNs)

Quantum Neural Networks replace classical neurons with parameterized quantum circuits.

Key Concepts:

- Quantum Gates as Activation Functions.
- Variational Quantum Circuits (VQCs) for optimization.

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Parameterized Quantum Circuit:

$$U(\theta) = \prod_i R_y(\theta_i) \cdot CNOT \cdot R_x(\theta_i)$$

Advantage: - Quantum gradients enable exploration of non-convex landscapes.

3. Quantum Boltzmann Machines (QBMs)

Quantum Boltzmann Machines leverage quantum mechanics to sample from a probability distribution.

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Quantum Hamiltonian:

$$H = - \sum_i b_i \sigma_i^z - \sum_{ij} w_{ij} \sigma_i^z \sigma_j^z$$

Advantage: - Efficient sampling in complex probability distributions.

Why Quantum?

- **Quantum Parallelism:** Process multiple states simultaneously.
- **Quantum Entanglement:** Correlated states for richer information.
- **Quantum Interference:** Constructive and destructive interference to enhance solutions.

Quantum Speedups in ML

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Example - Grover's Algorithm:

Quantum Search Complexity: $O(\sqrt{N})$ vs. $O(N)$

Advantage: - Speedups in high-dimensional optimization and linear algebra problems.

Challenges and Limitations

1. Quantum Hardware Limitations:

- Noisy Intermediate-Scale Quantum (NISQ) devices.
- Decoherence and limited qubit coherence times.

2. Data Encoding:

- Efficient embedding of classical data into quantum states.

3. Scalability:

- Difficult to scale circuits to large datasets.

Applications of Quantum Machine Learning

1. Quantum Chemistry:

- Simulate molecular structures with QML.

2. Finance:

- Quantum optimization for portfolio management.

3. Image Recognition:

- Quantum-enhanced convolutional neural networks.

1. Fault-Tolerant Quantum Computing:

- Overcoming noise for stable quantum circuits.

2. Hybrid Quantum-Classical Models:

- Combining quantum circuits with classical neural networks.

3. Quantum Internet:

- Distributed quantum machine learning over quantum networks.

References

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